

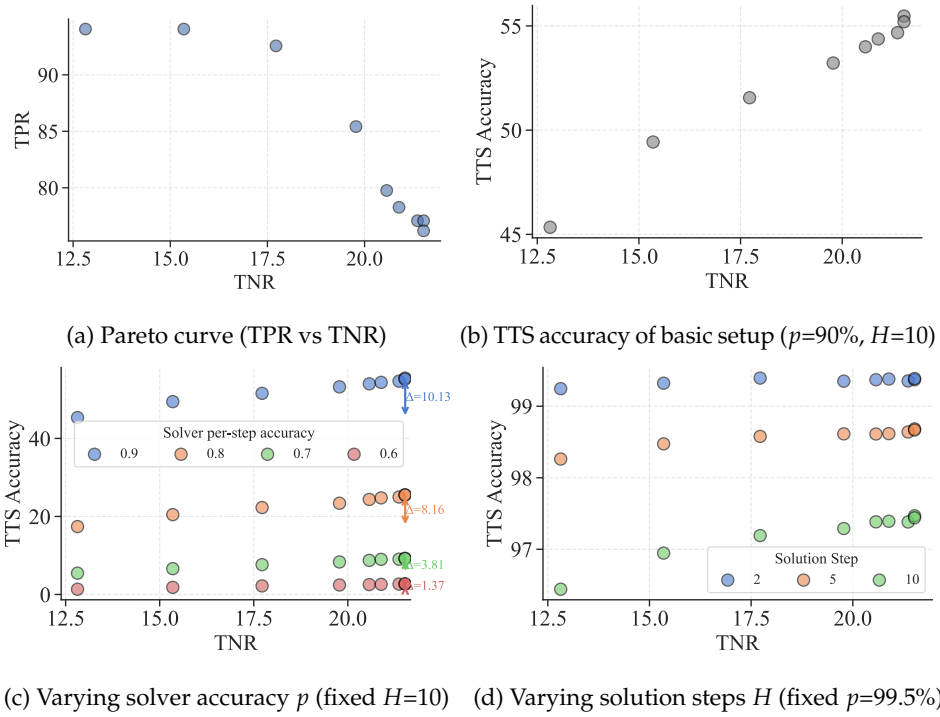


Figure 6: **Pareto curve of TPR vs. TNR and its impacts on performance of test-time scaling (TTS) method, localized rejection sampling.** Results shown for Qwen3-8B-thinking on Hard2Verify. (a) Varying the sample-level threshold τ produces large drop alongside an increase in TNR. (b) Simulated TTS accuracy under the basic setup ($p = 90\%$, $H = 10$): accuracy increases steadily as the verifier is steered toward higher TNR, despite the simultaneous drop in TPR. (c) Higher TNR yields higher TTS accuracy across all solver accuracy levels. (d) Higher TNR is slightly less beneficial when the solution is short ($H=2$) and solver accuracy is high; otherwise it consistently improves accuracy.